

ENGINEERING REPORT

A REPORT ON THE RESULTS OF
CONDUCTING AN OBJECTIVE AND
SUBJECTIVE TEST ON THE HYPERCUBE
AND JBL L-40 LOUD-SPEAKER ENCLOSURES

JANUARY, 1978

KESSLER ASSOCIATES
TELECOMMUNICATIONS CONSULTING ENGINEERS
GAINESVILLE, FLORIDA

INTRODUCTION

This report documents the results of comparative measurements to establish the relative performance characteristics of the HYPERCUBE loudspeaker enclosure and the JBL L-40 enclosure. Two sets of observations were made:

- (a) Comparative measurements with a sound-level meter to establish objectively the relative low-frequency response of the HYPERCUBE enclosure with respect to the JBL L-40 enclosure.
- (b) A series of subjective A-B comparisons under identical acoustical environmental conditions to assess the relative performance of the HYPERCUBE and the JBL L-40 enclosures.

OBJECTIVE MEASUREMENTS OF RELATIVE LOW-FREQUENCY PERFORMANCE

Measurement Method

The relative low-frequency response characteristics of the HYPERCUBE and the JBL L-40 enclosure was determined by measuring the response of each under identical electrical and environmental conditions using an audio signal generator, a power amplifier, and a sound pressure meter. The measurements were made on the grounds of WUFT-TV which is characterized by a large open area to ensure the absence of acoustical reflections and a sufficiently low ambient voice level so that the radiation from each speaker would not be contaminated by spurious sounds. The sound pressure meter and integral microphone assembly was placed on the surface of the ground approximately six feet away from the enclosure undergoing test to eliminate all reflections below 500 Hz.

The first series of measurements were made with the JBL L-40 speaker system. The audio signal generator output was held constant over the frequency range from 10 Hz to 1,000 Hz, and the sound pressure as indicated by the sound pressure meter, were recorded. After the measurements with the JBL L-40 were completed, the identical measurement technique and conditions were used to measure the sound pressure response of the HYPERCUBE enclosure.

It is important to note that the HYPERCUBE enclosure was equipped with woofer, and tweeter driver systems identical to the driver system employed by the JBL L-40 loudspeaker system so that any difference in performance would be due solely to the characteristics of the enclosures.

All objective measurements of frequency response were limited to frequencies below 1,000 Hz since the frequency response performance characteristics

of a loud speaker system above 1,000 Hz is determined largely by the characteristics of the drivers rather than the enclosures. In this instance, the low-frequency response characteristics below 500 Hz is due largely to the characteristics of the enclosure. Furthermore, to ensure accurate measurements of relative performance of the HYPERCUBE and the JBL L-40 enclosures below 500 Hz, the sound pressure radiations from each speaker system were normalized to zero db at 1,000 Hz.

The subjective comparisons between the HYPERCUBE and JBL L-40 enclosures were made in a typical living room listening environment with the HYPERCUBE speaker placed approximately 10 feet apart and approximately 10 feet away from all walls so that the directional and stereo characteristics could be judged. The speakers were driven with a typical high quality stereo system using recorded program material that was familiar. Since only a single JBL L-40 speaker system was available, the subjective listening tests consisted of two parts:

- (a) A subjective comparison of a single L-40 with the HYPERCUBE under identical conditions and program material.
- (b) Listening tests of a pair of HYPERCUBE speakers with special emphasis on the low frequency transient response, directional radiation characteristics, and the absence or presence of a stereo "hole" between the HYPERCUBE loudspeakers.

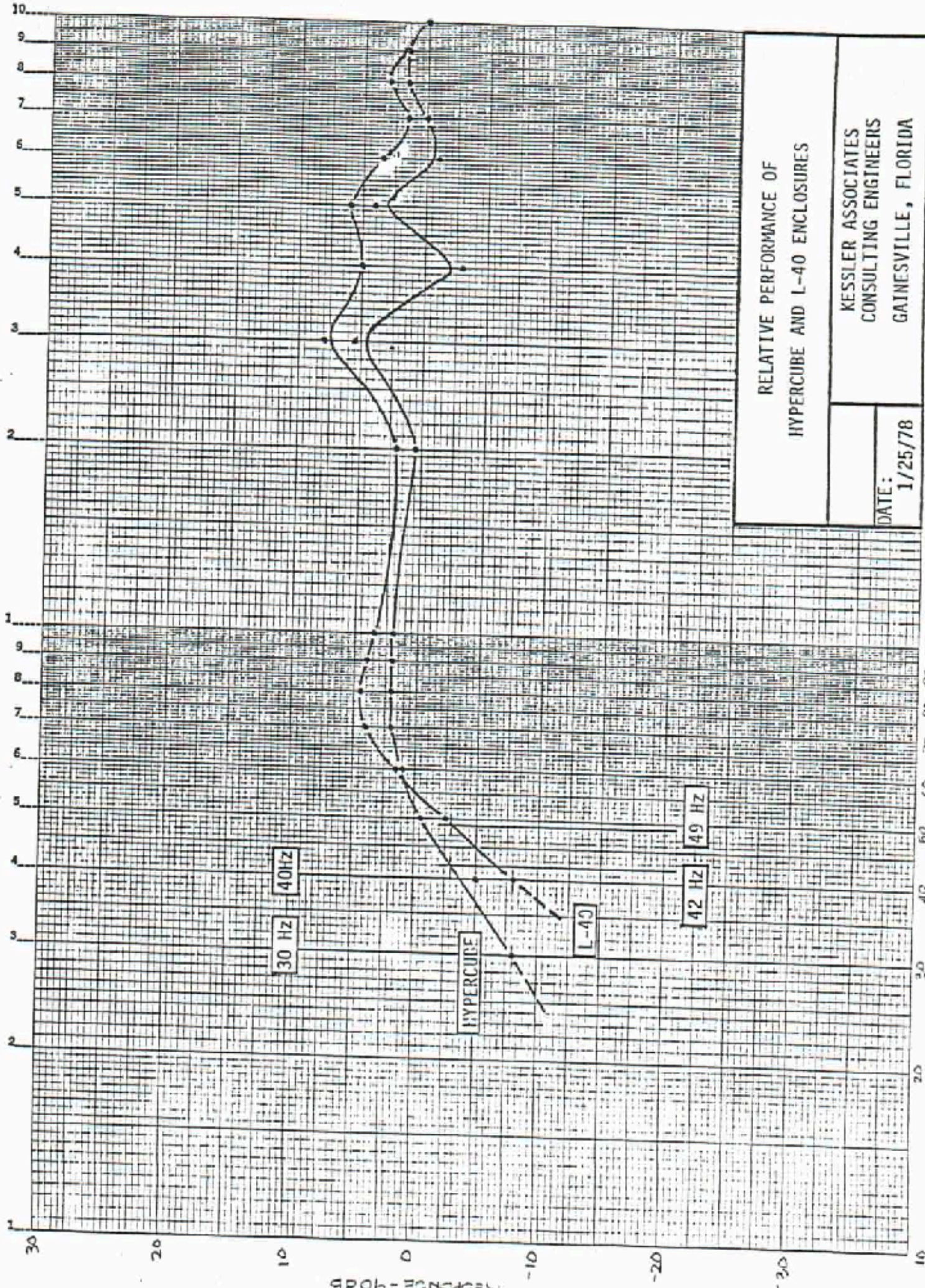
TEST RESULTS

The results of the measured comparison of the HYPERCUBE and L-40 low-frequency response characteristics is shown in the figure in which the measured low-frequency sound level radiations from each speaker are plotted. This figure is typical of at least eight sets of measurements. The only differences between the eight observations (which were not significantly large) occurred above 500 Hz and were due to constructive and destructive interference phenomena between the direct radiation from the speaker and pressure radiations reflected from the ground at the location of the sound pressure meter. This phenomena was minimized by placing the sound level meter on the surface of the ground. With this placement of the sound pressure meter, the undesirable effects of ground reflections below 500 Hz are virtually eliminated.

An examination of the figure discloses that the low-frequency performance of the HYPERCUBE speaker system, which appears to be due solely to the characteristics of the HYPERCUBE enclosure, is superior to the performance of the L-40 enclosure. This is demonstrated by the fact that the minus 3 db response frequency for the HYPERCUBE is 42 Hz while the minus 3 db response for the L-40 is only 49 Hz. Furthermore, since the measurements were conducted down to 10 Hz, the pressure response for the L-40 disappeared at 40 Hz, whereas the response of the HYPERCUBE disappeared at a frequency 10 Hz lower at a low-frequency of 30 Hz. This difference appears to be due solely to the low-frequency radiation characteristics of the enclosures since both enclosures employ identical drivers.

The results of the subjective listening tests appeared to indicate that the HYPERCUBE enclosure produced less "coloring" of the low-frequency transient characteristics of the reproduced music than the L-40 enclosure.

The subjective listening tests of the directional and stereo performance of the two HYPERCUBE enclosures revealed what this observer considers to be a remarkable non-directional characteristic. The apparent low-frequency



RELATIVE PERFORMANCE OF
HYPERCUBE AND L-40 ENCLOSURES

KESSLER ASSOCIATES
CONSULTING ENGINEERS
GAINESVILLE, FLORIDA

DATE:
1/25/78

sound level noted at the sides and rear of the two HYPERCUBE speakers appeared to be only slightly less than the sound level radiated in front of the speakers. This effect was noted as a result of walking around each speaker and around the pair of speakers and particularly noting the apparent change in sound level. Although the ear is not a very reliable device for judging absolute sound pressure level, it is quite accurate for judging relative sound pressure levels. The apparent non-directional characteristics of the HYPERCUBE also manifested itself during listening tests with particular attention devoted to the presence or absence of a stereo "hole" in between the two speakers. Walking in front of the speakers from one side to the other with the eyes closed failed to permit identification of a significant radiation "hole" between the speakers.

CONCLUDING COMMENTS

This observer frankly concedes that he can offer no explanation for the performance of the HYPERCUBE. However, it appears that two factors may be involved:

- (a) The irregular shape of the enclosure may inhibit the production of resonances of the back wave within the enclosure which would produce undesirable "coloration" of the front wave radiation;
- (b) the small rigid triangular surfaces of different dimensions with which the HYPERCUBE enclosure is fabricated may be serving as secondary radiators which reproduce non-resonant radiations essentially in phase with the radiation of the woofer cone.

Although the objective measurements of the comparative response of the HYPERCUBE and L-40 loudspeaker systems constitute a reliable measure of the relative low-frequency performance of the two enclosures, the subjective listening tests in a typical living room environment using recorded program material represent the subjective opinion of this observer only. The nature of subjective listening tests are such that each individual should listen for himself using familiar program material in a familiar playback environment and form a personal judgment.

 *William J. Kessler*
Kessler, P.E.